Best-of-N or Tree Search? A Controlled Study of Learned-Dynamics Optimism Under Test-Time Planning

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Abstract

Tree search over a learned dynamics model is usually framed as planning. This note studies a complementary view: search is also a test-time optimizer over the learned model, and can therefore amplify learned-model errors. We build a controlled continuous-control point-mass task with a localized optimistic "bias pocket" in the learned dynamics and reward surrogate. Static best-of-N evaluates independent rollouts, while MCTS adaptively expands and backs up branches that look valuable under the model. In a repeated-seed full run, UCT MCTS exhibits a larger selected-return optimism gap than best-of-N at the largest tested forward-model budget, and uncertainty-calibrated search reduces the gap below both. The contribution is not a robotics benchmark result; it is a compact mechanism testbed for studying when adaptive planning improves behavior and when it concentrates compute on model optimism.

1 Motivation

Learned model-based agents such as PlaNet [3], Dreamer [2], and PETS [1] demonstrate that learned dynamics can support efficient control. MCTS variants are also being extended to imperfect or continuous transition models [4, 5]. At the same time, recent test-time compute work has sharpened the distinction between static best-of-N sampling and adaptive search against a learned scorer [6].

This repository connects those views. If a learned dynamics model is also the test-time evaluator, then MCTS is an optimizer over a fallible model. More compute can improve planning, but it can also increase exposure to localized optimism. The goal is to make that effect measurable in a small setting before moving to larger robotics tasks.

2 Controlled Mechanism

The environment is a deterministic 2D point robot. The true reward encourages progress to a goal, penalizes action cost, and mildly penalizes an off-route pocket. The learned surrogate is accurate almost everywhere but, near that pocket, it predicts extra reward and a transition drift toward the goal. It also reports elevated uncertainty in that region.

For a selected action sequence $a_{0:H-1}$, we measure

$$\Delta_{\text{opt}} = \hat{G}(a_{0:H-1}) - G(a_{0:H-1}),$$

the learned-model return minus the true rollout return for the same sequence. We also record cumulative reward bias, transition error, uncertainty exposure, root visit concentration, and depth-wise bias.

3 Planners

The planner suite uses the same discrete action set and forward-model budget accounting:

- **Random open loop**: one random sequence.
- **Best-of-N**: independent model rollouts, selecting the maximum learned return.
- **UCT MCTS**: adaptive expansion with UCB selection and discounted backups.
- **Value MCTS**: MCTS with a learned leaf value estimate.
- **Uncertainty MCTS**: subtracts an uncertainty penalty during model scoring.
- **Conservative MCTS**: subtracts uncertainty during backup.

The intended repair is a lower-confidence search target. If uncertainty $u(s, a)$ upper-bounds local optimism, then replacing $\hat{r}$ with $\hat{r} - \lambda u$ in search or backup can suppress the branch before it dominates the root statistics.

4 Results

The full run uses 20 seeds and five budgets from 64 to 1024. The headline artifact is saved in `results/full/manifest.json`. At budget 1024, the selected-return optimism gap is larger for UCT MCTS than for static best-of-N, while the best calibrated repair reduces the gap:

Planner	Optimism gap at budget 1024	Root concentration
Best-of-N	0.301	n/a
UCT MCTS	0.686	0.649
Value MCTS	0.397	0.316
Uncertainty MCTS	0.236	0.727
Conservative MCTS	0.261	0.729

Table 1: Mean diagnostics from `results/full/summary.csv`. Lower optimism gap is better. Root concentration is the fraction of root visits assigned to the most visited child, defined for tree-search methods.

5 Interpretation

The result supports the narrow mechanism claim: in this controlled model, adaptive tree search can select trajectories whose learned return is more optimistic than the trajectories selected by static best-of-N under equal model-step budgets. The uncertainty-calibrated variants reduce the selected-return gap in the largest-budget run.

The result should not be overclaimed. The learned model is hand-designed rather than trained. The uncertainty signal is intentionally aligned with the injected bias pocket. The experiment is a microscope, not a benchmark. Its value is that the failure mode is observable, testable, and easy to extend.

6 Limitations and Next Steps

The next version should train an ensemble dynamics model on biased data, add continuous-action expansion, and evaluate on at least one standard continuous-control benchmark. The repair should be stress-tested under miscalibrated uncertainty and value-function optimism. The most important open question is whether search concentration predicts real downstream regret when the learned model is trained rather than hand-specified.

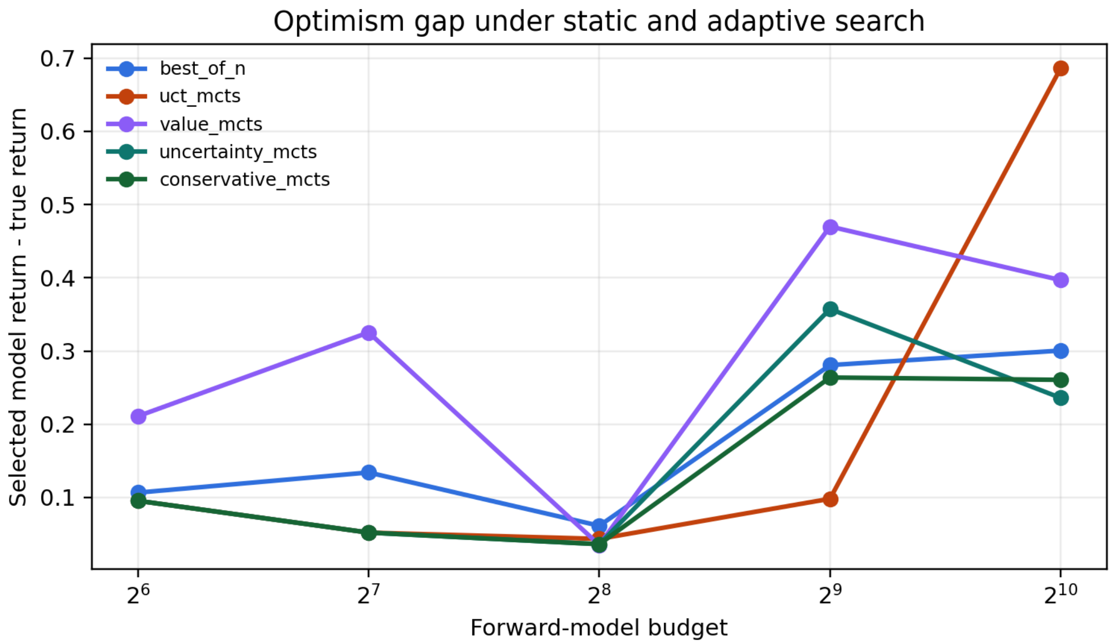

Figure 1: Selected-return optimism gap across forward-model budgets.

References

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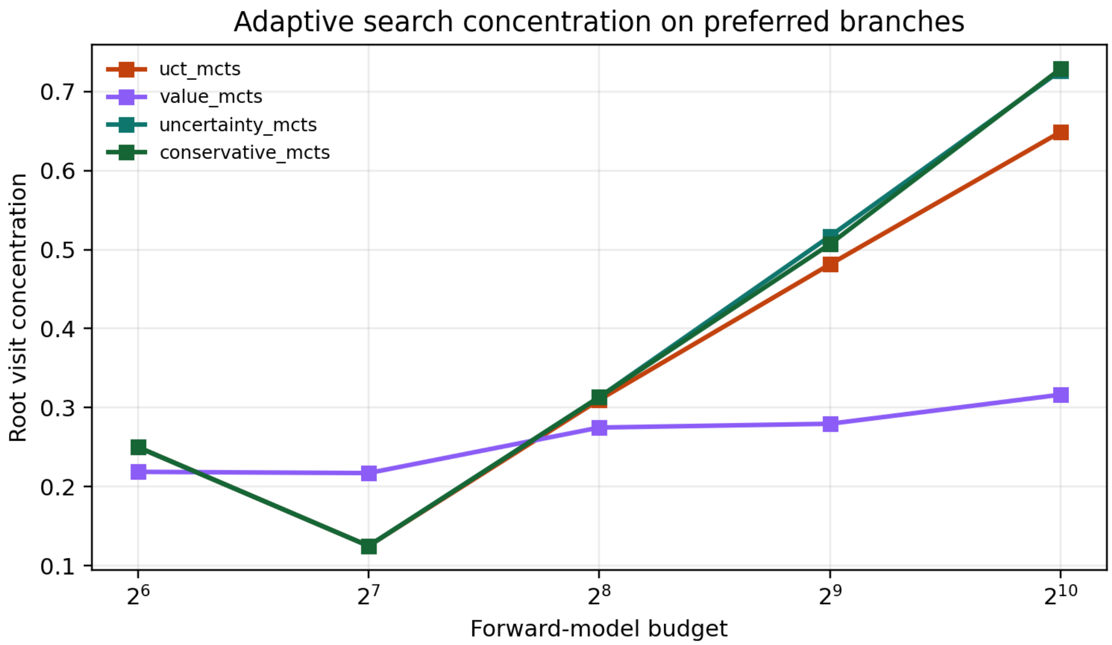

Figure 2: Root visit concentration increases as adaptive tree search allocates budget to preferred branches.

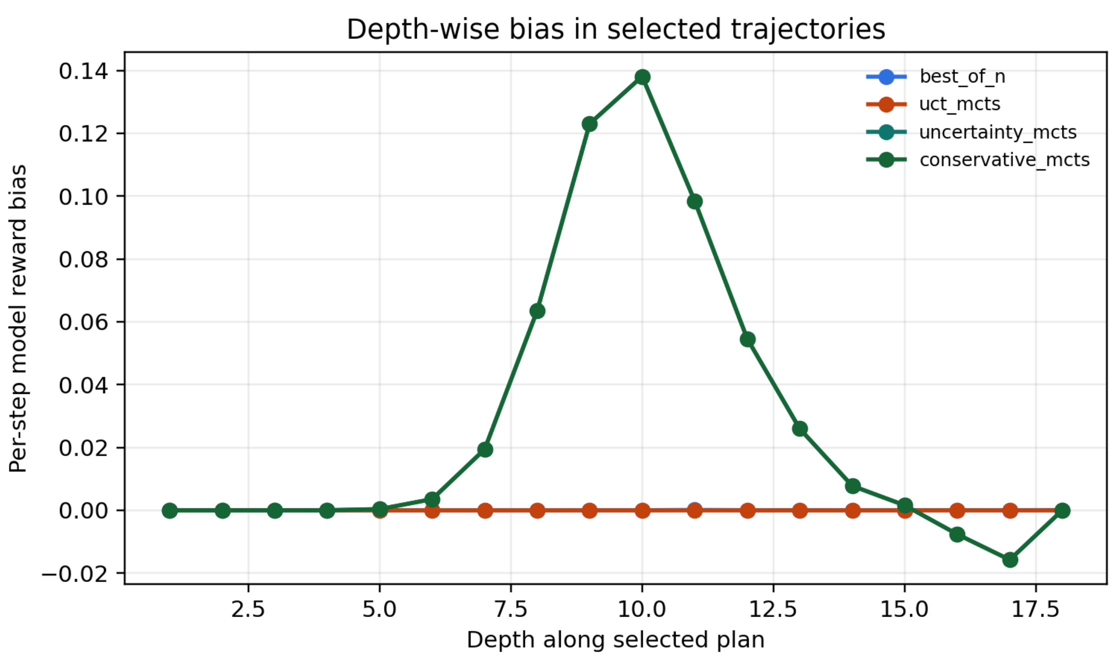

Figure 3: Depth-wise learned reward bias along selected plans in the largest-budget run.